A-A-54015
25 October 1989

# COMMERCIAL ITEM DESCRIPTION

## ELECTROCARDIOGRAPH

### Automatic, Solid State, Portable

This General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an all solid state, single channel, dual speed, direct writing, portable electrocardiograph with a minimum of three sensitivities.

Salient characteristics:

The device shall be all solid state, single channel, dual speed, direct writing, portable, with a minimum of three sensitivities. It shall have twelve lead recording capability for I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5 and V6 leads.

The strip chart recorder shall have two selectable speeds of 25 and 50 mm/sec, plus or minus 2 percent.

The electrocardiograph shall have sensitivities of 5, 10, and 20 mm/mV, plus or minus 2 percent.

The recorder shall have a thermal stylus or dot matrix printhead. If a stylus, it shall be easily replaceable, producing sharp, clear, and uniform traces. There shall be adjustments for stylus heat, position, and pressure. Hysteresis-return to baseline and baseline drift shall be minimal. For heated stylus recorders, the recording chart paper shall be 63 mm wide, overall 150 feet long, with a 50 mm grid width ruled in 1 mm by 5 mm squares. For thermal array (dot matrix) recorders, the recording chart paper shall be 59 mm wide, overall 150 feet long, with a 45 mm grid width ruled in 1 mm by 5 mm squares.

Manual and automatic recording modes shall be selectable for each lead.

Automatic lead marking shall function for both automatic and manual modes.

AMSC/NA FSC 6515

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Calibration pulse shall function for both automatic and manual modes.

The electrocardiograph shall be protected for standard Lown or equivalent differential pulse to 400 watt seconds terminated to patient.

Isolation circuitry shall limit total leakage current between patient leads (ECG) and ground to the value specified in ANSI/AAMI ES 1 - 1985, American National Standard, Safe Current Limits for Electromedical Apparatus.

Electrical characteristics. The equipment item furnished by the contractor must be capable of operating on both 50 and 60 Hertz (Hz) power sources. Voltage characteristics for this item shall permit the item to be operated on both of the following single phase voltage sources: 110 volt, 60 Hz; and 110 volt, 50 Hz. Items offered which normally operate on a nominal voltage range of 220 volt to 380 volt (single phase) rather than 110 volt will be accepted only if the item is capable of functioning on both 50 and 60 Hz. In order to meet this requirement, contractors may include a 60 Hz conversion kit with adequate instructions for the Government to perform the modification. The kit will be separately identified and priced on the bid/proposal and include a statement as to the number of manhours required to perform the modification. Complex modifications requiring more than one manhour to accomplish will not be accepted. In instances in which a 50 or a 60 Hz rated product is offered which does not require modification to allow operation at other than its rated frequency, the contractor will so certify in his bid/proposal, as well as identify any special operating conditions/limitations/requirements and liability and considerations which might ensue because of operation on other than the item's normal rated frequency.

Accessories. Each unit shall be supplied with the following standard accessories:

- One patient cable shall be supplied.
- One each adult Welsh suction chest electrode shall be supplied.
- One power cord shall be supplied.
- Four each adult limb electrodes shall be supplied.
- Four each adult limb straps shall be supplied.



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- One each tube/bottle electrode cream/gel shall be supplied.
- Two each rolls ECG chart paper shall be supplied.
- Chart paper mounts shall be supplied.
- A dust cover shall be supplied.

The electrocardiograph shall conform to the standards of electrical leakage currents for operator safety as specified in UL544 - "Standard for Safety of Electrical Medical and Dental Equipment", and NFPA 99-1987, Chapter 9, "Manufacturer Requirements."

Two complete instruction, operation, maintenance, and repair manuals shall be furnished.

The patient cable leads supplied with the electrocardiograph shall be color coded in conformance with AHA publication, "Recommendation for Standardization of Leads and of Specification for Instruments in Electrocardiography and Vectorcardiography".

The patient cable connector shall be a 3 lead/6 pin type configured in accordance with AAMI ECGC-5/83 "Standard for ECG Connectors".

The electrocardiograph performance standards shall be in accordance with ANSI/AAMI EC11-1982 "American National Standard for Diagnostic Electrocardiograph Devices".

The electrocardiograph shall measure no more than 20 inches long by 15 inches wide by 6 inches high.

The unit shall weigh no more than 10 pounds.

Workmanship. The electrocardiograph shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One electrocardiograph, as specified, constitutes one unit.



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## QUALITY ASSURANCE PROVISIONS

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one electrocardiograph, as specified in this description.



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Sampling for tests. Sampling for tests including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one electrocardiograph.

Tests.

Performance test. The electrocardiograph shall be completely assembled in its operating configuration and tested to determine compliance with this specification. Testing shall include Acceptance Testing.

Input dynamic range test.

The ECG device shall be capable of responding to and displaying differential voltages of plus or minus 5 millivolts (mV) varying at a rate up to 320 mV per second from a dc offset voltage in the range of -300 mV to +300 mV, when applied to any lead. The time-varying output signal amplitude shall not change by more than plus or minus 5 percent over the specified range of dc offset.

Time base accuracy test.

Record for at least 6 seconds at each time base, disregarding or discarding the first 1 second of data in each strip. Using calipers and a vernier scale calibrated to 0.1 mm, measure the distance between 10, 20, and 40 successive peaks; the distances must be within 10 plus or minus 0.2 mm, 20 plus or minus 0.4 mm, and 40 plus or minus 0.8 mm, respectively. Repeat the measurements at least three times at different parts of the strip for each time base. Verify that the measurements fall within the plus or minus 2 percent error band each time.

Metric products: Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.



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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE. The following National Stock Number is covered by this document.

6515-01-160-6006

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.



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Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DOD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4333

Location: CID/
ELECTROCARDIOGRAPH/H17-35



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| A-A-54015 | ELECTROCARDIOGRAPH |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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